

Ben Nguyen

(239) 888-0663 | benn5334@gmail.com | github.com/HuuBen5334 | linkedin.com/in/nguyenhuuben | Portfolio

EDUCATION

University of Florida

B.S. Computer Engineering – GPA: 3.65

Gainesville, FL

Expected May 2027

Study Abroad: Cross-Cultural Engineering Design, Kyoto University (Fall 2025)

Coursework: Data Structures & Algorithms, Microprocessor Applications, Digital Logic, Circuits I, AI for Virtual Environments

TECHNICAL SKILLS

Languages: C, C++, C#, Java, Python, JavaScript, AVR Assembly, VHDL, SQL

Embedded & Hardware: ESP32, Arduino, Altium Designer (PCB), I2S, Quartus Prime, Onshape CAD, 3D Printing

Software & Frameworks: Spring Boot, React, gRPC, REST APIs, WebSockets, PostgreSQL, Godot, Git

Certificates: Siemens Expedite Microcredential, AutoCAD Inventor

PROJECTS

Portable Bluetooth Speaker – *ESP32, C++, Altium Designer, I2S, SolidWorks*

- Designed a custom PCB in Altium Designer integrating dual MAX98357A class-D amplifiers with a ESP32 module, routing I2S, OLED, and control nets with decoupling and a ground pour to isolate switching noise from the audio path.
- Developed C++ firmware (Arduino framework) streaming Bluetooth audio over the I2S digital protocol to dual 5W speakers, with hardware interrupts and debouncing for tactile play/pause push-buttons.
- Programmed an OLED display to render real-time system status and playback metadata; modeled and 3D-printed an enclosure to house the PCB, speakers, display, and controls.

Full-Stack E-Commerce Marketplace – *Java 21, Spring Boot, C++, gRPC, React, PostgreSQL*

GitHub Repo

- Architected a four-tier distributed marketplace: Spring Boot 3.5 REST API, React 19 storefront, PostgreSQL database, and a multi-threaded C++ dynamic-pricing microservice invoked over gRPC.
- Eliminated a ~96% order-failure rate under a 50-user concurrent Locust load test by implementing optimistic locking (@Version) with automatic retry and backoff, plus foreign-key indexing on hot write paths.
- Delivered real-time order notifications via STOMP-over-WebSocket, replacing client polling; validated the system with Mockito unit tests, controller slice tests, and an end-to-end WebSocket integration test.

Robocode Tournament AI – *Java, Robocode API, Finite State Machines, Fuzzy Logic*

GitHub Repo

- Achieved 1st place among 26 competitors by developing a hybrid finite-state-machine and fuzzy-logic robot tank AI that outscored the benchmark bot by 91%.
- Utilized event-driven programming with the Robocode API for radar scanning, predictive targeting, and movement control.

FishSchooling Simulation – *C#, Godot Engine, Git*

GitHub Repo

- Implemented Reynolds flocking (alignment, cohesion, separation) on an interface-based steering system to model autonomous 2D agents; collaborated with 3 developers via Git branching and pull requests.
- Built a dynamic UI exposing fish-type-specific parameter sliders, enabling real-time behavior tuning across 5 creature types in a predator-prey ecosystem.

EXPERIENCE

UF Association of Computer Engineers

Gainesville, FL

Workshop Lead

March 2026 – Present

- Improved club engagement by hosting research lab tours and technical workshops; coordinated with presenters and staff to ensure seamless event execution.

Kyoto University

Kyoto, Japan

Networking Event Coordinator

September 2025 – December 2025

- Organized lunch networking events averaging 60 attendees with structured conversation activities to facilitate cross-cultural student interaction.
- Increased Japanese student attendance by 40% over prior semesters through faculty outreach and campus-wide promotion.

Blackwood Tennis Management

Sanibel / Fort Myers, FL

Customer Service Associate & Court Maintenance Specialist

May 2023 – August 2024

- Managed daily shop operations (transactions, check-ins, court reservations) and resolved customer inquiries.